



Noise-robust oversampling for imbalanced data classification

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ABSTRACT

The class imbalance problem is characterized by an unequal data distribution in which majority classes have a greater number of data samples than minority classes. Oversampling methods generate samples for minority classes to balance the data distribution. However, the generated minority samples may overlap with majority samples, resulting in noise. In this paper, we propose a noise-robust oversampling algorithm for mixed-type and multi-class imbalanced data. Our proposed noise-robust designs include an algorithm to eliminate noise within clusters of data samples, adaptive embedding to generate samples safely, and a safe boundary for enlarging class boundaries. The heterogeneous distance metric and adapted decomposition strategy render our noise-robust algorithm suitable for mixed-type and multi-class imbalanced data. Experimental results on 20 benchmark datasets demonstrate the effectiveness of the proposed algorithm.

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1. Introduction

Classification is an important pattern recognition task that attempts to assign each sample to a given class [1,2]. Imbalanced classification, as a specific classification type, occurs when one or some classes (the minority class) contain far fewer samples than the other classes (the majority class). The data distribution may be skewed in various application scenarios, such as fraud detection [3], disease diagnosis [4], and oil spill detection [5]. For example, the data used for disease diagnosis (e.g., mammography) include mostly normal samples and only a small proportion of abnormal samples [4]. When addressing this kind of data, traditional classifiers tend to ignore abnormal samples and classify all mammogram images as normal to achieve high classification accuracy. However, the abnormal samples that belong to the minority class are the most concerning.

Oversampling methods offer a promising means of alleviating the class imbalance problem by generating new samples for the minority class [6]. Yet most of oversampling methods are plagued by noisy sample generation [7,8]. Some noisy minority samples may overlap with the majority samples due to the improper generated features. In this paper, we identify several issues in the existing model designs that may cause the noisy sample generation. First, some scholars group the given data samples into multiple clusters, then generate samples within each discovered

cluster [9,10]. However, the performance of these methods is generally constrained by the adopted clustering algorithm. For instance, as shown in Fig. 1 (a), samples generated within the discovered cluster may contain noise because these samples are located in the majority region. In this case, the adopted clustering algorithm groups two dense clusters into one. Second, many oversampling studies have generated samples for a single minority sample based on its minority neighbors. However, as shown in Fig. 1 (b), some isolated (i.e., trapped) minority samples are far from their minority neighbors, causing the generated sample fall into the majority region. Third, some oversampling methods generate samples between the minorities and their majority neighbors to expand minority regions. In such instances, synthetic samples fill the empty space between classes and create unclear class boundaries, as illustrated in Fig. 1(c). Therefore, a noise-robust algorithm is expected to alleviate above issues: the improper clustering algorithm, unsafe trapped sample generation, and unclear boundaries between majority and minority classes.

Beyond model designs, existing imbalanced data solutions are also limited in terms of application scenarios. First, current solutions are mainly proposed for numerical variables [10,11]. However, mixed-type data including numerical and categorical variables are common in everyday life [12]. It is hard to adapt numerical solutions to categorical variables, as the selected distance metrics cannot always handle categorical variables appropriately [9]. Another application scenario limitation is that current solutions are originally proposed for the binary imbalanced classification. There is still a lack of research on the multi-class imbalance problem. Some studies decompose the multi-class problem into several bi-

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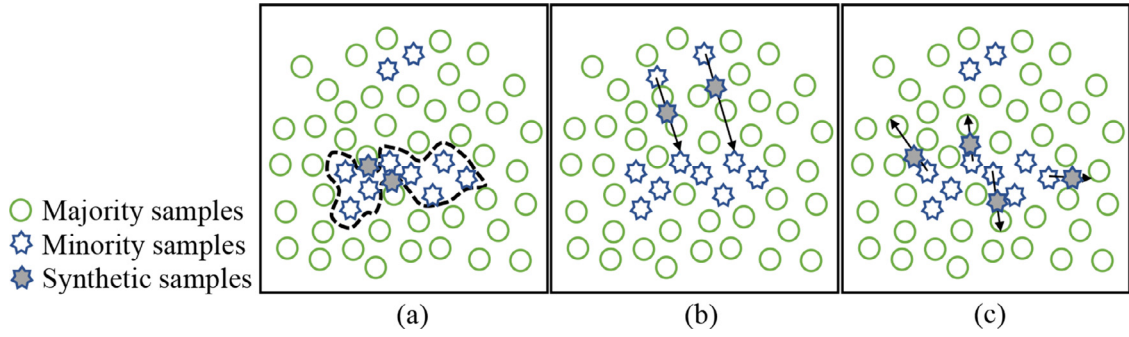


Fig. 1. Illustration of issues in noisy sample generation. (a) Synthetic samples within the cluster (b) Synthetic samples around isolated minorities (c) Synthetic samples between classes.

nary problems, then apply existing binary data solutions [10,13]. But these approaches may lose useful information (e.g., between-class relationships) [13]. Especially, the noisy sample generation issue may be more severe in the multi-class case since data samples of different classes may overlap in multiple regions.

This paper proposes a Noise-Robust Oversampling for imbalanced data classification, called NROMM. NROMM largely mitigates noisy sample generation and can deal with the mixed-type and multi-class data. The model designs of NROMM include an algorithm to eliminate noise within the clusters of data samples, an adaptive embedding algorithm to generate samples more safely, and a safe boundary to enlarge the boundaries between classes.

The rest of this paper is organized as follows. Related work is introduced in Section 2. The noise-robust model designs are described in Section 3. Application scenario extensions for mixed-type and multi-class data are detailed in Section 4. Section 5 presents the experimental setting and corresponding results. Finally, we conclude this paper and outline future research directions in Section 6.

2. Related work

2.1. Clustering methods

Clustering algorithms partition data samples into groups so the samples in each group are alike [14]. K-means is one of the most widely used approaches, partitioning all samples into K groups based on similarity [14]. Nevertheless, this algorithm is biased toward majorities, as it aims to minimize the within-cluster sum of squares [14]. A density-based clustering algorithm NBDOS was recently proposed to uncover arbitrary cluster shapes [10]. This technique calculates each data sample's density as the proportion of samples with the same class label among the k -nearest neighbors. If a sample's density is not less than the threshold, then the sample and its neighbors are grouped into the same cluster. Yet NBDOS is relatively ineffective when separating clusters if data are densely distributed and adjacent to each other. In addition, this algorithm cannot identify any cluster if the samples in each class are sparsely interspersed with those in other classes [9].

The CFSFDP algorithm [15] has been proposed to overcome the limitations of NBDOS [10] by adding a constraint to cluster centers; that is, the centers are relatively far apart. The detailed clustering process is as follows:

- Given f -dimensional vector representations of n samples $\mathbf{X} \in \mathbb{R}^{n \times f}$, CFSFDP calculates the mutual distance as $D = [d_{ij}]_{n \times n}$, where d_{ij} denotes the distance between $\mathbf{x}_i$ with $\mathbf{x}_j$.
- Then $\mathbf{x}_i$'s local density ρ_i and the minimum distance from $\mathbf{x}_i$ to any other sample with a higher density δ_i are calculated as:

$$\rho_i = \sum_{j: j \neq i} \exp \left[- \left(\frac{d_{ij}}{d_c} \right)^2 \right], \quad \delta_i = \min_{j: \rho_j > \rho_i} (d_{ij}), \quad (1)$$

where $\exp$ is the exponential function; and d_c is the distance threshold, empirically the first 2% value after sorting d_{ij} in ascending order [15].

- The samples with simultaneously high ρ_i and δ_i represent centers, and the remaining samples are assigned to the nearest center.

CFSFDP recognizes those samples that have higher densities than their neighbors and are relatively far from samples with higher densities as centers. Therefore, CFSFDP will not group two dense clusters into one based on a relatively large distance between centers.

2.2. More application scenarios of the imbalanced classification problem

2.2.1. Binary vs. multi-class imbalanced problems

The class imbalance problem is often studied in binary settings. However, this issue also arises in multi-class scenarios [16]. Binarization solutions are often used to decompose the multi-class problem into several binary problems. Each binary sub-problem can thus directly apply to existing solutions. For example, enhanced automatic twin support vector machine [16] adopts two binarization solutions one-versus-one (OVO) and one-versus-rest (OVR) to extend this algorithm to the multi-class setting. However, directly adopting either OVO or OVR would limit multi-class extension performance. OVO only preserves two classes at a time and loses information about the remaining classes [13]. Multi-class radial-based oversampling (MC-RBO) [17] represents another decomposition strategy (one-versus-sampling [OVS]) to reduce information loss. In every binary sub-problem, the majority class is formed by sampling observations from larger classes.

2.2.2. Numerical vs. categorical variables

Generally, it is hard to extend most of existing imbalanced classification works from the numerical case to categorical case because of the adopted distance metrics. Euclidean, Manhattan, and Chebyshev distances are three common metrics for measuring differences between numerical variables [18]. Euclidean distance is calculated by adding up the square root of coordinate differences. Manhattan distance is computed by adding up the absolute coordinate differences. Chebyshev distance is the maximum value among all absolute coordinate differences.

One-hot encoding is the most popular means of encoding each categorical value with 0 or 1 [19]. Another categorical distance metric is the value difference metric (VDM) [18]. VDM assumes that two categorical values are closer if they have a similar output class. However, if there is only one output class, then the VDM for all categorical values should be 0.

The entropy method can be adopted [20] to calculate differences among categorical variables within each class. Assume that the categorical variable a has M_a categorical values in total. $P(a_m)$

is the probability of the m -th categorical value a_m occurring. The entropy weight H_a is defined as $H_a = -\frac{1}{M_a} \sum_{m=1}^{M_a} P(a_m) \log P(a_m)$. The averaged entropy weight is defined as $w_a = H_a / \sum_{s=1}^{f_c} H_s$, where f_c is the total number of categorical variables. The distance metric $d(\cdot)$ is defined as $d(\mathbf{x}, \mathbf{y}) = \sum_{a=1}^{f_c} w_a \cdot \mathbb{1}(\mathbf{x}_a, \mathbf{y}_a)$, where $\mathbb{1}(x) = 1$ if $x \neq 0$ and $\mathbb{1}(x) = 0$ otherwise.

2.3. Oversampling methods

Oversampling methods replicate or generate samples to balance the class distribution [6]. Interpolation-based oversampling methods, such as SMOTE [11], generate a sample $\mathbf{s} \in \mathbb{R}^f$ of the given minority sample $\mathbf{x} \in \mathbb{R}^f$ by linearly interpolating between $\mathbf{x}$ and the minority neighbor $\mathbf{x}'$. SMOTE adopts the Euclidean metric to define the neighborhood. However, this distance measure is unsuitable for high-dimensional data [21]. Feature-weighted oversampling (FW-SMOTE) has accordingly been proposed which assigns different feature weights while calculating the distance between samples [21]. Still, FW-SMOTE follows SMOTE in using interpolation to generate samples. When some isolated minorities are far from their minority neighbors, SMOTE and FW-SMOTE each easily generate samples in the majority region.

As stated in Section 1, some scholars have combined clustering algorithms with oversampling methods to reduce noise. Potential anchoring (PA) [22] adopts the K-means algorithm to cluster all samples into K partitions. New samples are then generated based on density measurements of the K partitions. Position-aware safe boundary interpolation oversampling (PABIO) [23] adopts the CFSFDP clustering algorithm to cluster minorities into an inland set I , a borderline set B , and a trapped set T . Moreover, it proposes the idea of safe boundaries in that samples are generated around trapped minorities. More recently, an oversampling algorithm called SVDDWSMOT [24] has leveraged the CFSFDP clustering algorithm to detect candidate noisy minorities that are characterized by lower densities and greater relative distances. It also adapts the one-class classification method, SVDD, to detect and remove overlapped samples around class boundaries.

While the above clustering-based oversampling algorithms alleviate the class imbalance problem, their utility remains limited for binary and numerical classification tasks. The class imbalance problem also applies under multi-class scenarios for mixed-type data. Position-aware interpolation oversampling (PAIO) [9] adopts the VDM metric to find candidate minority neighbors while interpolating samples. MC-RBO [8] uses the radial basis function to estimate the potential of every sample in the multi-class setting. Multi-class combined cleaning and resampling (MC-CCR) [13] defines an energy function to estimate the optimal radius of sphere regions around every minority sample. All three methods alleviate the class imbalance problem in more complicated cases. However, they still have constraints in solving the imbalance problem for mixed-type and multi-class data.

To explain oversampling methods more fully, we compare them on the bases of technical mechanisms and application scenarios in Table 1. The generation issues related to PAIO [9] reveal three mechanisms that may result in noise. First, PAIO adopts an improper clustering algorithm. As discussed in Section 2.1, NBDOS [10] cannot separate adjacent clusters in dense regions. This inability leads to a synthetic sample within the cluster in the majority region (Fig. 2(a)). The second risk of noise generation accompanies an improper approach to generate samples around minorities trapped in the majority region. PAIO generates samples in the majority region, although it filters out far-off minority neighbors to generate samples around trapped samples (Fig. 2(b)). Third, PAIO lacks a cleaning strategy, leading to an unclear class boundary between the minority and majority classes (Fig. 2(c)). These issues are expected to be rectified.

Advanced work remains fairly limited in terms of application scenarios. As stated in Section 2.2.2, the VDM metric cannot measure categorical differences within the same class. One-hot encoding increases the number of binary variables and results in a bias toward categorical variables [9]. We thus propose adapting a distance metric to handle heterogeneous data that include numerical and categorical variables. A dedicated distance metric should be adopted in both clustering algorithms (e.g., CFSFDP) to calculate the distance between samples and in the interpolation process to find candidate neighbors. Additionally, we propose adapting the binarization solution to avoid information loss in the multi-class problem. Instead of using sub-information about classes, we aim to integrate all information from classes larger than the current oversampled class.

3. Noise-robust oversampling algorithm

As discussed in Section 1 and 2, existing algorithms are likely to generate noisy samples due to the improper clustering algorithm, unsafe trapped sample generation, and unclear boundaries among classes. In this section, we first summarize our NROMM for binary and numerical imbalanced data classification and then detail three key improvements.

3.1. Overflow

Our algorithm comprises two main phases: *oversampling minority samples* and *cleaning majority samples*. The overflow is summarized in Alg. 1 and visualized in Fig. 3. More specifically, we propose three noise-robust model designs: adapting CFSFDP [15] to avoid generating noisy samples, developing an adaptive embedding algorithm to generate samples more safely, and proposing a safe boundary to enlarge the boundary between classes.

- As noted in Section 2.3, the deficiencies of NBDOS [9] can lead to noise. Accordingly, we adapt CFSFDP [15] in Step 2 of Alg. 1, (see Section 3.2).
- The trapped samples T easily generate noise around them. We therefore propose adaptive embedding for S_T in Step 3 of Alg. 1 (see Section 3.3).
- As an unclear class boundary can hinder classifiers, we suggest cleaning the majority samples in Step 4 - 6 of Alg. 1 (see Section 3.4).

3.2. Adapt CFSFDP algorithm and divide the minority samples

In this subsection, we display recommended updates to Step 2 in Alg. 1. As indicated in Section 2.3, PAIO suffers from noise generation for inland samples because two dense clusters may be

Algorithm 1 Noise-robust oversampling algorithm for binary and numerical imbalanced data classification.

Require: The imbalanced dataset X .

- 1: Separate X into the minorities $X^{\min}$ and the majorities X^{maj} ;
Oversample the minority samples $X^{\min}$
 - 2: Employ the *clustering algorithm* and divide $X^{\min}$ into inland I , borderline B , trapped T ;
 - 3: Interpolate to generate S_I, S_B ; adaptive embedding for S_T ;
Clean the majority samples X^{maj}
 - 4: Employ the *clustering algorithm* and divide X^{maj} into inland I , borderline B , trapped T ;
 - 5: Remove the majority borderline B and trapped samples T ;
 - 6: The cleaned majority samples: $\tilde{X}^{\text{maj}} \leftarrow X^{\text{maj}} \setminus \{B \cup T\}$.
 - 7: **return** $\tilde{X} \leftarrow X^{\min} \cup S_I \cup S_B \cup S_T \cup \tilde{X}^{\text{maj}}$
-

Table 1

The overall comparison of oversampling algorithms. “Clustering” measures whether the algorithm integrates the clustering algorithm. “Oversampling trapped” measures how samples are generated around minority samples trapped in the majority region. “Cleaning” measures whether the algorithm cleans up the samples that hinder classifiers.

Algorithms	Technical Mechanisms			Application Scenarios			
	Clustering	Oversamplingtrapped	Cleaning	Binary	Multi-class	Numerical	Categorical
SMOTE [11]	×	×	×	✓	×	✓	×
FW-SMOTE [21]	×	×	×	✓	×	✓	×
PA [22]	K-means [14]	×	×	✓	×	✓	×
SVDDWSMOTE [24]	CFSFDP [15]	×	✓	✓	×	✓	×
PABIO [23]	CFSFDP [15]	×	✓	✓	×	✓	×
PAIO [9]	NBDOS [10]	interpolation	×	✓	×	✓	VDM [18]
MC-RBO [17]	×	×	✓	✓	OVS [17]	✓	One-hot [19]
MC-CCR [13]	×	×	✓	✓	OVS [17]	✓	One-hot [19]
NROMM (Ours)	CFSFDP [15]	adaptive embedding & interpolation	✓	✓	OVS	✓	Entropy [20] & VDM [18]

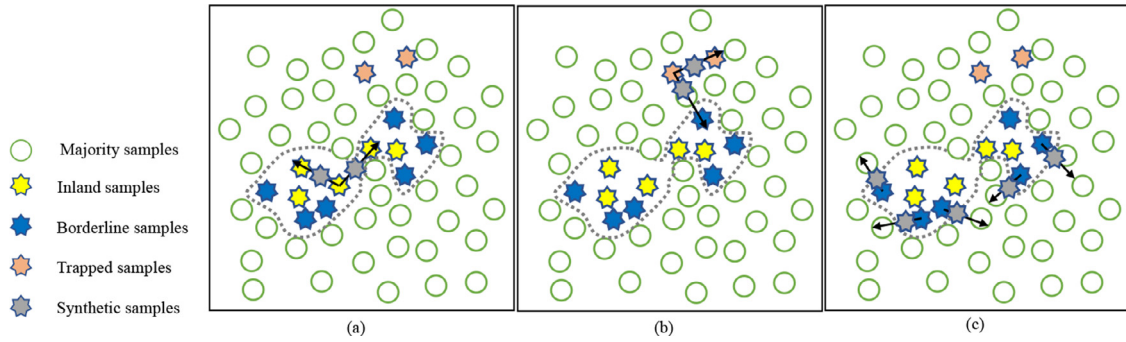


Fig. 2. Illustration of sample generation and issues in PAIO. (a) Synthetic samples around I reside in the majority region (b) Synthetic samples around T reside in the majority region (c) Synthetic samples around B make the class boundary unclear.

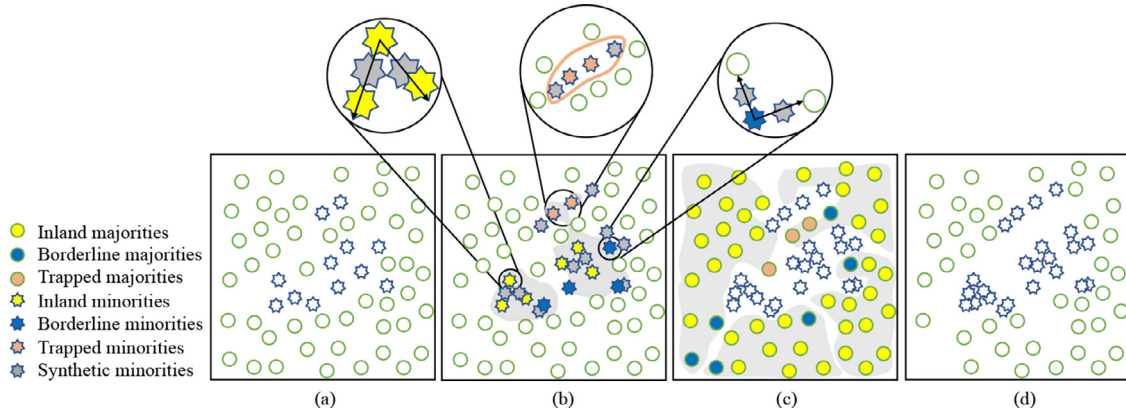


Fig. 3. Illustration of the proposed algorithm with binary imbalanced data. From the left: (a) Original dataset (b) Discover two clusters on $X^{\min}$; divide $X^{\min}$ into I , B , T , and generate S_I , S_B , S_T (c) Discover four clusters on X^{maj} and divide X^{maj} into I , B , T (d) Remove B , T from X^{maj} ; ready for classification.

grouped into one. We propose using CFSFDP clustering [15] to address this problem. CFSFDP limits the relative distance between cluster centers (see Section 2.1) to prevent two dense clusters from being grouped together. However, CFSFDP [15] only performs clustering; it cannot be directly applied in our algorithm because it does not distinguish sample types within each cluster. We instead adapt the CFSFDP algorithm with our proposed division method.

Given the minority samples $X^{\min} \in \mathbb{R}^{n_{\min} \times f}$, we apply CFSFDP [15] to group $X^{\min}$ into clusters $L = \{L_1, \dots, L_{n_L}\}$, where $L_m \subset X^{\min}$ and $L_i \cap L_j = \emptyset$ for any $i \neq j$. For a minority sample $x_i \in L_m$, let $N_{\min}(x_i)$ denote its k_1 -nearest neighbors in $X^{\min}$ while $N_{\min}^*(x_i) = \{x_j | x_j \in N_{\min}(x_i) \cap L_m\}$ denotes its k_1 -nearest neighbors in the cluster. The local density $\rho(x_i)$ of x_i is calculated as $\rho(x_i) = |N_{\min}^*(x_i)|/k_1$. Based on $\rho(x_i)$, we divide $X^{\min}$ with a density threshold ρTh (Fig. 3(b)):

- **Inland sample:** A sample x_i is characterized by $\rho(x_i) > \rho Th$.
- **Borderline sample:** A sample x_i is characterized by $\rho(x_i) \leq \rho Th$, and an inland sample exists in its k_1 -neighborhood.
- **Trapped sample:** A sample x_i is characterized by $\rho(x_i) \leq \rho Th$, and not a single inland sample exists in its k_1 -neighborhood.

3.3. Adaptive embedding for trapped samples

In this subsection, we illustrate suggested updates to Step 3 in Alg. 1. As shown in Section 2.3, existing algorithms generate samples only based on minority neighbors, resulting in noise around trapped samples. To mitigate this issue, we propose an adaptive embedding algorithm that adopts majority information to safely generate samples for trapped samples. Synthetic samples should not be in the majority region to avoid creating noise and to main-

tain a certain distance from their trapped sample neighbors to maintain diversity. As such, our first hypothesis is as follows:

Hypothesis 1 (SYNTHETIC ASSOCIATION). The synthetic samples should be more similar (i.e., closer in the embedding space) to their corresponding trapped samples than to any other majority sample.

Given a trapped sample t_i , its affiliated trapped subset is $T_i = \{t_i, t_j | t_j \in N(t_i) \cap L_m\}$, where $N(t_i)$ denotes the neighbor set of each t_i and L_m is the cluster label of t_i . Correspondingly, there are k_2 -nearest neighbors in X^{maj} , denoted as $N_{maj}(t_i)$. Then the set of majority neighbors around T_i is $M_i = \cup_j N_{maj}(t_j)$ for all $t_j \in T_i$. Lastly, all trapped samples are denoted as $T = \cup_i T_i$, and all majority neighbors around T are denoted as $M = \cup_i M_i$.

Given the number of synthetic samples that need to be generated for each trapped sample $\zeta(t_i)$, S_{T_i} represents the synthetic samples for each t_i , where $|S_{T_i}| = \zeta(t_i)$. All synthetic samples for the trapped samples T are denoted as $S_T = \cup_i S_{T_i}$. To discern the embedding vector of each s_j , we extend the margin-based loss in [25] and define a partial loss σ_j^{Ass} as follows:

$$\sigma_j^{Ass} = \max \left\{ 0, 1 - \left[\max_{t \in T_i} \phi(\mathbf{t}, \mathbf{s}_j) - \max_{\mathbf{m} \in M_i} \phi(\mathbf{m}, \mathbf{s}_j) \right] \right\}, \quad (2)$$

where $\phi(\mathbf{a}, \mathbf{b})$ represents the normalized Euclidean distance between vectors $\mathbf{a}$ and $\mathbf{b}$. Intuitively, for each synthetic sample, the maximum similarity score associated with its nearest trapped neighbors T_i should be greater than the maximum similarity score associated with any other majority neighbor M_i .

However, only optimizing σ_j^{Ass} cannot satisfy our expected oversampling results because doing so solely generates samples that overlap with the trapped sample. We therefore put forth another hypothesis to maintain the diversity between synthetic samples.

Hypothesis 2 (SYNTHETIC DIVERSITY). The synthetic sample should be relatively far from its trapped neighbor set in the embedding space.

For every synthetic sample s_j and its affiliated trapped neighbor set T_j , $\bar{t}_j$ denotes the mean of T_j . We define a partial loss σ_j^{Div} to maintain the relative distance between s_j and T_j as follows:

$$\sigma_j^{Div} = \frac{1}{1 - \cos(\bar{\mathbf{t}}_j, \mathbf{s}_j)}, \quad (3)$$

where $\cos(\mathbf{a}, \mathbf{b}) = \frac{\mathbf{a}^T \mathbf{b}}{|\mathbf{a}| |\mathbf{b}|}$ represents the cosine similarity between vectors $\mathbf{a}$ and $\mathbf{b}$. Minimizing σ_j^{Div} forces the synthetic sample to be relatively far from its nearest trapped neighbor set.

We next combine two component losses (Eq. 2 & Eq. 3) into an overall objective function:

$$\sigma_j^{Total} = \alpha \cdot \sigma_j^{Ass} + \beta \cdot \sigma_j^{Div}, \quad (4)$$

where α and β are two parameters to balance the two loss terms. We adopt the alternative minimization method [25] to update T , M , and S_T . The premise behind adaptive embedding for the synthetic samples around trapped samples is shown in Fig. 3(b).

3.4. Clean majority samples near boundary

In this subsection, we detail the proposed updates to Step 4 - 6 in Algorithm 1. As discussed in Section 2.3, the class boundary becomes unclear if borderline minorities are oversampled based on their majority neighbors. We recommend using the clustering algorithm to discover borderline and trapped majorities and then remove them. Given majority samples $\mathbf{X}^{maj} \in \mathbb{R}^{n_{maj} \times f}$, we apply CFSFDP [15] to group $\mathbf{X}^{maj}$ into clusters $L = \{L_1, \dots, L_{n_L}\}$, where $L_m \subset X^{\min}$ and $L_i \cap L_j = \emptyset$ for any $i \neq j$. For a majority sample $x_i \in$

L_m , $N_{maj}(x_i)$ denotes its k_3 -nearest neighbors in X^{maj} , and $N^*(x_i) = \{x_j | x_i, x_j \in N_{maj}(x_i) \cap L_m\}$ denotes its k_3 -nearest neighbors in its cluster. The local density of x_i is calculated as $\rho(x_i) = |N^*(x_i)|/k_3$.

Similar to the division of minorities in Section 3.2, we divide the majority samples with their local densities in Fig. 3(c). The borderline and trapped majority samples both have relatively low local densities: $\rho(x_i) \leq \rho_{Th}$. After that, we propose removing the borderline and trapped majorities to enlarge the boundary between classes in Fig. 3(d).

4. Application scenario extensions

4.1. Handle mixed-type data

Real-world datasets often include a mixture of numerical and categorical variables [12]. To handle mixed-type data, we upgrade this algorithm from two perspectives: the adopted distance metric for clustering (Step 2 & 4) and sample generation with categorical variables (Step 3).

4.1.1. Heterogeneous distance metric in clustering

Unlike numerical variables, categorical variables do not have size-based relationships [9]. For instance, the categorical value "3" is not necessarily larger than "1". Relatedly, the distance between "3" and "1" is not greater than that between "2" and "1". We use the entropy distance function [20] for categorical variables and the Euclidean distance metric [18] for numerical variables. Given the data $\mathbf{X} \in \mathbb{R}^{n \times f}$ with both numerical and categorical variables, the heterogeneous Euclidean-entropy metric (HEEM) between two specific vectors $\mathbf{x}$ and $\mathbf{x}'$ is defined as follows:

$$HEEM(\mathbf{x}, \mathbf{x}') = \sqrt{\sum_{a=1}^f [d_a(\mathbf{x}_a, \mathbf{x}'_a)]^2}, \quad (5)$$

where f is the number of variables. The function $d_a(\mathbf{x}_a, \mathbf{x}'_a)$ returns the distance between two values $\mathbf{x}_a$ and $\mathbf{x}'_a$ of variable a , which is defined as:

$$d_a(\mathbf{x}_a, \mathbf{x}'_a) = \begin{cases} (|\mathbf{x}_a - \mathbf{x}'_a|)/(4\sigma_a), & \text{if } a \text{ is numerical} \\ w_a \cdot \mathbb{1}(\mathbf{x}_a, \mathbf{x}'_a), & \text{if } a \text{ is categorical} \\ 1, & \text{if either } \mathbf{x}_a \text{ or } \mathbf{x}'_a \text{ is unknown} \end{cases} \quad (6)$$

where σ_a is the standard deviation of variable a . To avoid bias toward either numerical variables or categorical variables, we normalize the Euclidean distance and the entropy weight distance to the same range $[0, 1]$.

4.1.2. Generate samples with categorical variables

Inspired by [9], we employ the interpolation method to generate S_I and S_B in Step 3 (Alg. 1). However, this approach will generate samples with noisy categorical variables. For example, the value interpolated between encoded values "3" and "1" is randomly in the range $[1, 3]$. 1.2 indicates no real categorical value. We thus propose a dedicated method to generate samples with categorical variables.

We apply the VDM [18] to calculate categorical value differences in the interpolation method inspired by [9]. As the synthetic sample s should be among the minority sample x and its neighbor x' , the difference of s from both x and x' should be smaller than the difference between x and x' . The intermediate value of each categorical variable a for sample s should hold to the constraints: $vdm_a(\mathbf{x}_a, \mathbf{s}_a) \leq vdm_a(\mathbf{x}_a, \mathbf{x}'_a)$ & $vdm_a(\mathbf{s}_a, \mathbf{x}'_a) \leq vdm_a(\mathbf{x}_a, \mathbf{x}'_a)$, where $\mathbf{s}_a$, $\mathbf{x}_a$, and $\mathbf{x}'_a$ are the values of variable a for synthetic sample s , minority sample x , and its neighbor x' , respectively. We propose adding another constraint to ensure that the synthetic sample is closer to its reference minority sample than to its neighbor

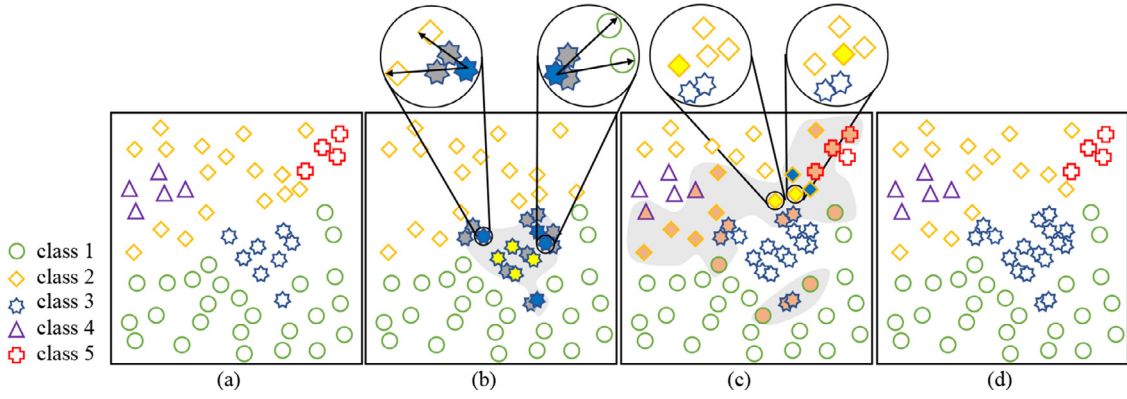


Fig. 4. Illustration of the proposed algorithm in the multi-class setting. From the left: (a) Original dataset, with X^3 being oversampled (b) Temporarily remove samples from smaller classes X^4 & X^5 , cluster and divide X^3 into I (yellow) and B (blue), and generate new samples for X^3 (c) Discover the overlapping set R and then cluster and divide the samples in R into I (yellow), B (blue), T (orange) (d) Remove the inland samples I from the overlapping set R .

sample: $vdm_a(\mathbf{x}_a, \mathbf{s}_a) \leq vdm_a(\mathbf{s}_a, \mathbf{x}'_a)$. Intermediate categorical values can be identified based on the above difference constraints. This algorithm is detailed in the supplementary materials.

4.2. Handle multi-class imbalanced data

As discussed in Section 2.2.1, between-class relationships and the class-overlapping problem are especially evident in the multi-class setting. Therefore, we upgrade Alg. 1 from two perspectives: oversampling for each class (Step 1) and cleaning the overlapped regions (Step 4 - 6). The procedures of our proposed algorithm in the multi-class setting appear in Fig. 4.

4.2.1. Oversampling for each class

For multi-class imbalanced data, it is challenging to decide whether to oversample or undersample on every class [13]. We binarize the multi-class problem and then oversample each class except for that with the maximum observations, $X^{\min}$, one at a time in the binary setting. Meanwhile, we collect the corresponding majority samples X^{maj} to generate S_B in Step 3. Inspired by [17], we propose combining the original and synthetic samples from classes larger than the class being oversampled as the majorities X^{maj} (one-versus-ensemble, [OVE]), as shown in Fig. 4(b).

4.2.2. Clean samples in overlapped regions

We present a dedicated cleaning approach for multi-class imbalanced data, as some classes would be cleaned repeatedly if using the binary cleaning strategy. Therefore, we recommend cleaning the samples in overlapping regions between classes once all classes have been oversampled.

Starting with an arbitrary sample $x_i \in X^c$, if its k_4 -nearest neighbor set $N(x_i)$ has samples from the other classes $\bar{X}^c$, then a class-overlapping set is formed, defined as $R = \{x_i, x_j | x_j \in N(x_i) \cap \bar{X}^c\}$. Next, we expand the overlapping set R until all samples have been examined. After that, we leverage CFSFDP [15] to group samples in R into clusters L . For a sample x_i that is in the overlapping set and assigned to cluster L_m , we denote the set of its k_5 -nearest neighbors that share the same class label c as $N_c(x_i)$. $N_c^*(x_i) = \{x_j | x_j \in N_c(x_i) \cap L_m\}$ denotes its nearest neighbors in the cluster. The local density of x_i is then calculated as $\rho(x_i) = |N_c^*(x_i)|/k_5$.

Similar to Section 3.2, we divide samples in the overlapping set R into inland, borderline, and trapped samples. Different from removing borderline and trapped majority samples under the binary cleaning strategy, we remove inland samples in the overlapping set (Fig. 4(c)). These inland samples are interior samples with the highest degree of overlap.

5. Experimental study

5.1. Experimental setting

Inspired by [8,13], we use the same 20 imbalanced datasets from the KEEL repository [12] as our experimental data. These datasets are summarized in Table 2. The imbalance ratio (IR) is computed by dividing the number of samples from the largest class by the number of samples from the smallest class. Class overlapping ($^\circ R$) is our proposed data characteristic, measuring the proportion of samples in the overlapping set R to all data samples. Specifically, we discover the degree of class overlap R with the 4-nearest neighbors. $\%I$, $\%B$, $\%T$ respectively denote the mean percentage of inland, borderline, and trapped samples among all classes.

We follow the experimental settings in MC-CCR [13] and MC-RBO [17]. Due to space limitations, detailed settings (including hyper-parameters) are presented in the supplementary material. Our NROMM is compared with the common baselines in MC-CCR [13] and MC-RBO [17]: SMOTE [11] using the round-robin decomposition strategy (SMOTE-all); STATIC-SMOTE (S-SMOTE) [26]; Mahalanobis distance oversampling (MDO) [27]; SMOTE combined with an iterative partitioning filter (SMOTE-IPF) [7]; Multi-class synthetic minority oversampling (SMOM) [10]. In addition, we compare NROMM with an advanced oversampling algorithm: K-nearest neighbor oversampling (KNNOR) [28]. To implement it on the multi-class data, we treat the largest class as the majority class and the rest of the classes as minority classes.

5.2. Comparison with reference oversampling algorithms

Due to space limitations, complete experimental results appear in the supplementary materials, including the classification performance of our NROMM against eight baselines for twenty benchmarks on four classifiers and four assessment metrics. We also present the comparison performance of our NROMM against Evolutionary Mahalanobis Distance Oversampling (EMDO) [29] in the supplementary material. We present the results under C5.0 decision tree classifier in Table 3. In this table, the results of all baselines except KNNOR are referred from [13,17]. Results of the Friedman ranking test [30] and Shaffer's post hoc analysis [30] are listed in Table 4. In Fig. 5, we visualize the results of the Wilcoxon signed-ranks test [31].

Performance of adapting CFSFDP Improper clustering algorithm would result in the noise generation, especially for the inland samples within clusters. In Section 3.2, we adopt CFSFDP [15] to close clusters and avoid the noise generation. The Page-blocks dataset

Table 2
Detailed description of multi-class imbalanced datasets used in experiments.

Data	#N	#C	#F _n	#F _c	Class distribution	IR	°R	%I	%B	%T
Wine	178	3	13	0	59/71/48	1.48	22.23	79.63	14.37	6.00
Glass	214	6	9	0	70/76/17/13/9/29	8.44	80.37	29.80	23.77	46.43
New-thyroid	215	3	5	0	150/35/30	5.00	28.97	50.27	27.27	22.50
Vehicle	846	4	18	0	199/212/217/218	1.17	69.74	35.50	39.83	24.68
Winequality-red	1599	6	11	0	10/53/681/638/199/18	68.10	87.98	17.73	31.07	51.18
Page-blocks	5472	5	10	0	4913/329/28/87/115	175.46	9.87	74.08	17.10	8.84
Zoo	101	7	0	17	41/13/10/20/8/5/4	10.25	36.00	48.57	44.29	7.14
Lymphography	148	4	0	18	2/81/61/4	40.50	59.46	40.83	7.73	51.45
Hayes-roth	160	3	0	4	65/64/31	2.10	90.00	23.40	39.23	37.37
Dermatology	358	6	0	33	111/60/71/48/48/20	5.55	34.78	63.63	30.53	5.85
Led7digit	500	10	0	7	45/37/51/57/52/52/47/57/53/49	1.54	67.20	59.27	8.50	32.23
Balance	625	3	0	4	288/49/288	5.88	63.78	33.57	31.93	34.50
Flare	1066	6	0	10	331/239/211/147/95/43	7.70	54.22	54.40	25.95	18.03
Car	1728	4	0	6	65/69/384/1210	18.61	39.47	40.65	42.23	17.10
Automobile	150	6	10	15	3/20/48/46/29/13	16.00	63.78	15.83	18.13	66.05
Cleveland	297	5	5	8	164/55/36/35/13	12.62	80.41	12.50	3.50	84.00
Ecoli	336	8	5	2	143/77/2/2/35/20/5/52	71.50	73.21	33.51	24.35	42.13
Contraceptive	1473	3	6	3	629/333/511	1.89	92.39	23.40	30.47	46.13
Yeast	1484	10	6	2	244/429/463/44/51/163/35/30/20/5	92.60	94.88	30.29	36.72	32.97
Thyroid	7200	3	6	15	166/368/6666	40.16	26.55	78.90	13.33	7.77

Note: #N the number of samples; #C the number of classes; #F_n the number of numerical variables; #F_c denote the number of categorical variables.

Table 3
NROMM and reference methods for some experimental data on C5.0 classifier.

Data Sets	Measurements	NROMM	MC-CCR	MC-RBO	SMOTE-all	S-SMOTE	MDO	SMOM	SMOTE-IPF	KNNOR
Page-blocks	AvAcc	90.90	<u>83.71</u>	<u>83.71</u>	75.83	75.25	78.47	77.56	74.20	85.73
	MG	88.20	82.55	<u>87.27</u>	71.64	74.38	79.06	77.24	76.72	83.47
	CBA	89.83	84.63	<u>86.73</u>	78.60	71.77	80.84	79.96	76.35	84.95
	CEN	0.09	<u>0.13</u>	0.15	0.34	0.29	0.24	0.19	0.26	0.10
Automobile	AvAcc	92.20	76.98	77.11	<u>80.12</u>	73.53	78.13	79.04	75.32	76.51
	MG	91.78	75.68	77.20	51.86	74.40	75.11	<u>78.75</u>	73.28	75.39
	CBA	87.40	77.93	<u>80.48</u>	54.47	71.79	75.11	73.35	70.84	72.77
	CEN	0.13	0.25	<u>0.23</u>	0.56	0.29	0.29	0.30	0.29	0.29
Cleveland	AvAcc	38.59	<u>37.88</u>	35.01	28.92	27.18	28.92	28.01	24.98	37.16
	MG	36.02	30.98	<u>34.04</u>	18.00	24.57	26.45	24.63	23.39	33.68
	CBA	36.41	33.91	35.01	24.09	25.44	27.34	27.34	21.99	<u>35.11</u>
	CEN	0.57	<u>0.62</u>	0.65	0.87	0.76	0.75	0.76	0.71	0.60
Hayes-roth	AvAcc	92.70	<u>92.11</u>	<u>91.88</u>	86.45	88.04	87.33	90.06	89.74	90.65
	MG	91.16	<u>85.16</u>	81.93	70.52	84.56	83.18	86.82	84.20	<u>89.92</u>
	CBA	91.73	<u>90.03</u>	88.64	77.48	86.30	84.96	87.62	75.52	88.34
	CEN	<u>0.16</u>	<u>0.16</u>	0.19	0.34	0.19	0.19	0.14	0.35	0.20
Zoo	AvAcc	97.14	85.92	<u>88.56</u>	82.61	68.69	79.09	79.09	67.30	81.46
	MG	96.86	84.29	<u>85.01</u>	55.61	63.47	75.83	77.65	78.55	78.88
	CBA	93.57	81.52	<u>82.51</u>	60.83	66.08	76.72	76.72	79.05	79.03
	CEN	0.02	0.19	0.21	0.48	0.38	0.27	0.27	0.22	<u>0.08</u>
Lymphography	AvAcc	89.08	79.6	<u>81.22</u>	73.02	62.67	76.54	74.72	74.20	75.15
	MG	88.22	77.36	<u>79.13</u>	40.60	58.32	73.08	73.99	71.86	72.98
	CBA	86.09	<u>80.66</u>	79.02	43.21	59.19	75.75	78.39	70.83	74.99
	CEN	0.20	<u>0.26</u>	0.26	0.68	0.43	0.31	0.34	0.40	0.35
Thyroid	AvAcc	<u>96.08</u>	80.52	89.43	80.02	85.34	79.14	80.96	78.91	97.64
	MG	<u>96.05</u>	81.48	86.94	73.62	81.86	75.88	77.70	75.10	97.61
	CBA	<u>94.94</u>	81.99	84.11	80.58	81.86	76.77	75.01	70.46	94.98
	CEN	<u>0.03</u>	0.17	0.09	0.34	0.22	0.27	0.29	0.29	0.02

Note: The best results are highlighted in **bold**, and the second-best results are underlined.

contains 74.08% of the inland samples (Table 2). For Page-blocks, our proposed algorithm achieves four of the best results on the four metrics (Table 3).

Performance of adaptive embedding The trapped samples are far from their minority neighbors, and the generated samples around them easily overlap with majorities. In Section 3.3, we propose adaptive embedding to generate samples more safely. The Automobile and Cleveland datasets contain 66.05% and 84.00% of the trapped samples, respectively (Table 2). Our NROMM achieves four of the best results on the four metrics (Table 3) for both of these datasets.

Performance of categorical variables Noisy variable values would be generated without the dedicated distance metric for cat-

egorical variables. In Section 4.1, we extend our model designs to the mixed-type data. The Zoo dataset has 17 categorical variables; with this dataset, NROMM achieves four of the best results on the four metrics.

Performance of safe boundary The noise generation issue may be aggravated in multi-class case since a sample may overlap with multiple classes simultaneously. In Section 4.2.2, we suggest identifying the multiple overlapping regions and cleaning up inland samples in them. Tab. 2 indicates that Hayes-roth has a class-overlapping degree of 90. For Hayes-roth, NROMM achieves three of the best results (Tab. 3).

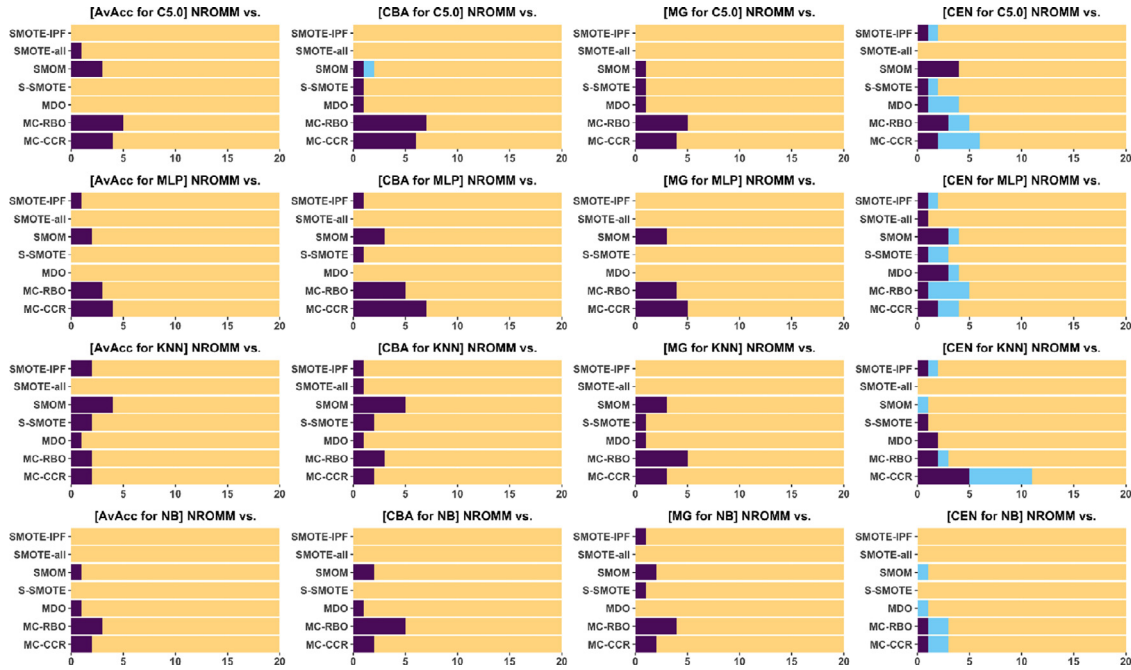
Performance of other data characteristics In Section 4.2, we extend our noise-robust oversampling algorithm to the multi-class

Table 4

Mean ranks of evaluated algorithms. The classification results of baselines that differ significantly from NROMM (Shaffers post hoc test) are denoted by '*'.

Algorithms	C5.0				MLP			
	AvAcc	CBA	MG	CEN	AvAcc	CBA	MG	CEN
SMOTE-all [11]	6.175*	8.550*	9.000*	8.950*	8.050*	8.400*	8.750*	8.250*
S-SMOTE [26]	6.850*	6.600*	6.050*	6.150 *	6.250*	6.650*	6.750*	6.600*
MDO [27]	5.675*	5.075*	5.475*	5.475*	6.575*	6.850*	6.925*	5.525*
SMOM [10]	4.800*	4.900*	4.800*	5.025*	4.850*	3.450*	4.400*	3.975*
SMOTE-IPF [7]	7.850*	7.450*	6.775*	6.525*	6.000*	6.850*	6.275*	6.125*
MC-CCR [13]	4.025*	3.450	3.800*	3.525	3.075	2.950	2.800	3.950*
MC-RBO [17]	3.225	2.450	2.900	3.125	4.200*	3.250	3.175	4.200*
KNNOR [28]	4.650*	4.600*	4.500*	4.200*	4.400*	4.750*	4.175*	4.350*
NROMM	1.750	1.925	1.700	2.025	1.600	1.850	1.750	2.025

Algorithms	KNN				NB			
	AvAcc	CBA	MG	CEN	AvAcc	CBA	MG	CEN
SMOTE-all [11]	8.450*	8.300*	7.750*	7.625*	8.100*	8.275*	8.125*	7.825*
S-SMOTE [26]	5.350*	6.175*	6.025*	7.550*	5.800*	6.075*	6.250*	7.375*
MDO [27]	7.425*	6.450*	6.450*	5.550*	7.250*	6.000*	6.800*	6.250*
SMOM [10]	4.600*	4.375*	5.200*	5.800*	4.900*	5.200*	4.775*	4.700*
SMOTE-IPF [7]	5.625*	6.650*	5.825*	5.300*	6.750*	6.350*	6.700*	6.925*
MC-CCR [13]	3.750*	3.700*	3.450	2.075	3.200*	3.900*	3.600*	3.000
MC-RBO [17]	3.950*	3.300	4.000*	4.625*	3.000	3.150*	3.000	3.250*
KNNOR [28]	4.100*	4.150*	4.500*	4.650 *	4.600*	4.450*	4.150*	4.275*
NROMM	1.750	1.900	1.800	1.825	1.400	1.600	1.600	1.400

Note: The best mean ranks are highlighted in **bold**. The significance level is 0.05.**Fig. 5.** Comparison results of the proposed NROMM with all eight baselines for all four classifiers and assessment metrics, including the number of datasets on which NROMM performs significantly better (yellow), statistically equal (blue), or worse (purple).

setting. We further illustrate its performance with respect to three data characteristics.

- 1) The Lymphography dataset has **small disjuncts**, which consists of two classes respectively containing only two and four samples (Table 2). Therefore, the class information for small disjuncts are rare. In Section 3.3, we use majority class information to learn the adaptive embedding for generating samples. Our proposed NROMM achieves four of the best results on the four metrics (Tab. 3) in this case.
- 2) The Page-blocks dataset has the **highest IR** (Tab. 2). Our proposed NROMM outperforms the other algorithms significantly on all four metrics for this dataset (Tab. 3).

- 3) Thyroid is the **largest dataset** among all 20 benchmarks. In this instance, NROMM achieves four second-best results on all four metrics (Tab. 3). Although KNNOR [28] achieves four of the best results on C5.0 classifier, it performs poorly on KNN and NB classifiers (Tab. 14 - 21 in the supplementary material).

In Table 4, we present the mean rank results of the Friedman test [31]. Our proposed NROMM shows the lowest mean ranks over all classifiers and assessment metrics. The average performance of NROMM is therefore demonstrably superior to all reference algorithms. Moreover, based on the results of Shaffers post hoc test [30], NROMM leads to significant differences on most classifiers and metrics compared with the baselines.

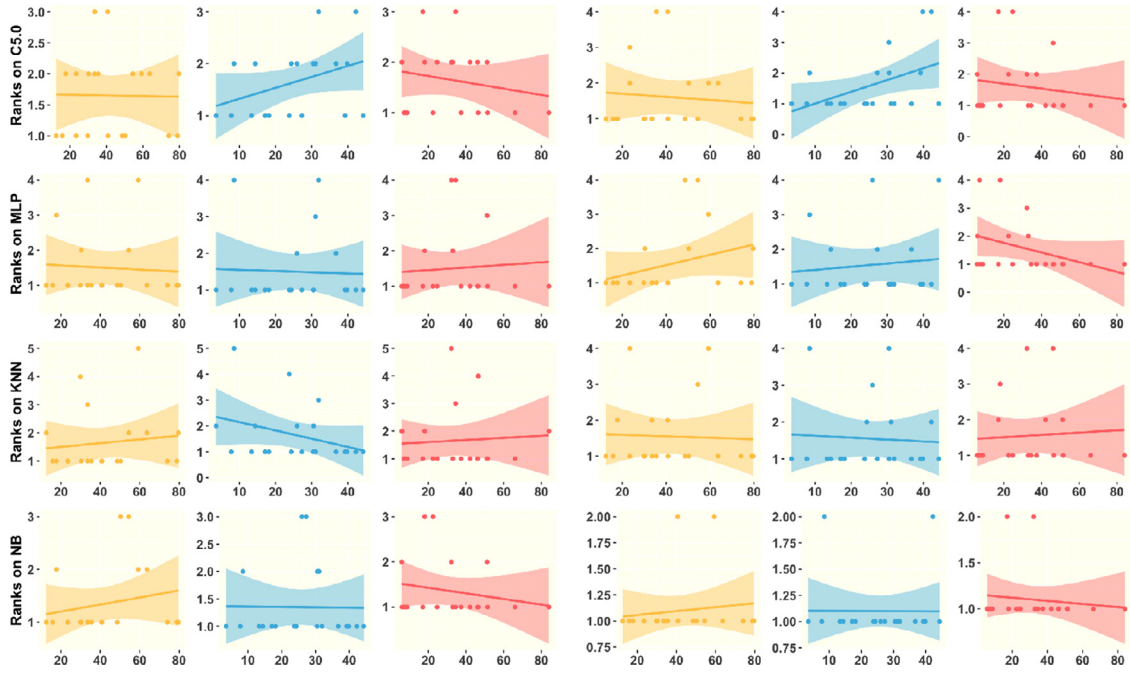


Fig. 6. Illustration of the linear relationship between the mean percentage of inland (yellow), borderline (blue), and trapped (red) samples across all classes and the proposed NROMM performance rank; left: AvAcc; right: CEN; each row for a classifier; fitted linear regression and 95% confidence levels are shown.

In Fig. 5, we present the number of datasets on which NROMM performs significantly better, statistically equal, or worse among the 20 datasets compared with each baseline. NROMM outperforms all reference oversampling algorithms on most datasets. Among the four classifiers, NROMM's performance on NB [13] is relatively better. Our algorithm performs similarly well to the AvAcc and CEN assessment metrics [17] when comparing the four metrics. The proposed algorithm thus improves classification accuracy, especially for the minority class, and helps alleviate misclassification.

5.3. Examination of validity of extension designs

First, we examine the validity of the design choices in our NROMM algorithm within the multi-class setting. The first variable in multi-class data classification is the cleaning strategy. We compare the dedicated approach to detect and remove inland samples in the overlapped region in Section 4.2.2 (cleaning, T) with the absence of such a strategy (cleaning, F). The second variable is the method for categorical variables, comparing the combination of HEEM in clustering and VDM [18] in interpolation (distance, H) with one-hot encoding [19] (distance, O). We compare our OVE decomposition strategy in Section 4.2 (decompose, E) with OVR [16] (decompose, R).

For experimental validation of our noise-robust designs, we compare all possible combinations of the outlined parameters on four classifiers and two assessment metrics. The averaged results, based on all benchmarks, are shown in Tab. 5. NROMM (containing the cleaning strategy, the combination of HEEM in clustering and VDM in interpolation, and the OVE decomposition strategy) displays the best performance on average. Specifically, the cleaning strategy is vital to achieving good results, such that the absence of this strategy leads to worse results.

In addition, we examine the validity of our noise-robust designs in binary classification by comparing our performance with the three most advanced oversampling algorithms: PA [22], SVD-DWSMOTE [24], and FW-SMOTE [21]. Due to length considerations, the 10 datasets from D1 to D10 are described in the supplemen-

tary materials. In Table 6, our noise-robust designs that include the cleaning strategy and the dedicated distance metric for categorical variables contribute to outstanding binary classification performance.

5.4. Effects of data characteristics

The forth stage of our experimental study involves analyzing the relationship between the percentage of each sample type and the rank of NROMM performance compared with baselines. Due to space limitations, we illustrate the performance on AvAcc and CEN assessment metrics [13] in Fig. 6. We also present the Pearson correlation coefficient [8] between the percentage of each sample type and the rank of NROMM performance on all metrics in Table 7. The ranking is in descending order, and the best performance is assigned Rank 1. The negative regression slope in Fig. 6 and the negative correlation in Tab. 7 show that our algorithm performs relatively better with a large percentage of the specific sample type.

For C5.0, our algorithm performs better on the four metrics for datasets containing a large percentage of inland and trapped samples but significantly worse on the four metrics for datasets containing a large percentage of borderline samples. For the classification performance of KNN on MG metric, our algorithm performs significantly better on datasets containing a large percentage of inland samples. No significant linear relationships are observed between the percentage of each sample type and our performance rank for other classifiers and assessment metrics. As such, our algorithm performs well when encountering an array of data characteristics.

5.5. Effect of oversampling ratio

Finally, we want to consider how oversampling ratios affect the oversampling algorithms performance. Fig. 7 depicts the impacts of oversampling ratios on the performance of the examined oversampling algorithms. NROMM outperforms other algorithms, especially for small oversampling ratios. Whereas other reference algorithms usually demonstrate optimal performance when the oversampling

Table 5
Impacts of NROMM parameters on assessment metrics and classifiers.

Parameters			C5.0		MLP		KNN		NB	
cleaning	distance	decompose	AvAcc	CBA	AvAcc	CBA	AvAcc	CBA	AvAcc	CBA
T	H	E	79.34	77.08	80.91	78.06	75.59	73.88	69.41	65.12
T	H	R	77.79	76.62	79.84	77.36	74.69	72.70	68.12	63.61
T	O	E	75.68	72.68	75.56	71.73	71.49	68.31	66.22	61.15
T	O	R	74.29	71.87	73.92	68.71	67.69	66.21	64.15	60.42
F	H	E	72.89	72.05	74.21	70.26	68.59	67.41	63.15	60.36
F	H	R	71.62	70.39	73.62	69.63	68.05	67.23	61.51	58.35
F	O	E	71.49	68.38	72.22	68.06	66.36	64.71	63.22	58.60
F	O	R	69.48	66.38	70.41	65.94	64.57	62.67	59.07	54.25

Note: The best results are highlighted in **bold**.

Table 6
Binary performance comparison for selected experimental data, classified by SVM on AUC and G-mean metrics.

		D1	D2	D3	D4	D5			D6	D7	D8	D9	D10
AUC	PA	0.912	0.785	0.817	0.885	0.884	AUC	PA	0.873	0.868	0.858	0.892	0.795
	SVDD	0.976	0.833	0.900	0.981	0.957		F-SMOTE	0.903	0.895	0.911	0.914	0.863
	Proposed	0.981	0.859	0.923	0.987	0.962		Proposed	0.905	0.953	0.931	0.932	0.893
GM	PA	0.911	0.761	0.815	0.884	0.881	GM	PA	0.867	0.861	0.852	0.890	0.782
	SVDD	0.935	0.745	0.837	0.907	0.867		F-SMOTE	0.879	0.819	0.802	0.895	0.871
	Proposed	0.942	0.821	0.860	0.913	0.872		Proposed	0.881	0.897	0.862	0.913	0.921

Table 7

Pearson correlation coefficient r between the percentage of the sample type and the rank of the NROMM algorithm with regard to the classifier and assessment metric.

	C5.0				MLP			
	AvAcc	CBA	MG	CEN	AvAcc	CBA	MG	CEN
Inland	-0.0149	-0.1903	-0.0716	-0.0911	-0.0615	-0.2739	0.1986	0.3117
Borderline	0.3797*	0.4551*	0.3826 *	0.4666*	-0.0394	0.2453	0.2015	0.1109
Trapped	-0.2008	-0.0697	-0.1513	-0.1716	0.0791	0.1221	-0.3017	-0.3714

	KNN				NB			
	AvAcc	CBA	MG	CEN	AvAcc	CBA	MG	CEN
Inland	0.1208	-0.0332	-0.3807*	-0.0419	0.2044	0.3294	0.0459	0.1267
Borderline	-0.3382	0.0750	-0.0767	-0.0621	-0.0134	-0.1186	-0.0738	-0.0043
Trapped	0.0722	-0.0079	0.4021*	0.0692	-0.1985	-0.2526	-0.0005	-0.1183

Note: '*' stands for the statistically significant correlation at the significance level of 0.10.

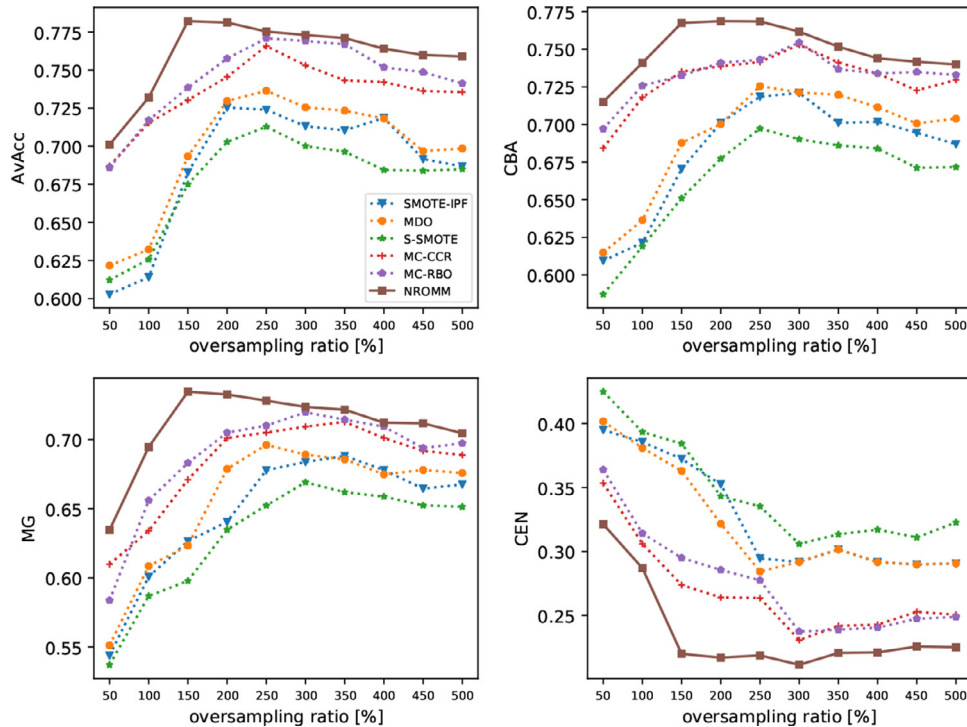


Fig. 7. Effects of varying oversampling ratios on the performance of six multi-class oversampling algorithms, averaged over 20 datasets, using the C5.0 classifier.

ratio ranges between 200% and 300%, our algorithm performs best when the ratio is 150%. Moreover, as the oversampling ratio continues to increase, the performance of all chosen algorithms tends to decline. This pattern indicates that when too many new samples are generated, the number of duplicate samples rises and exacerbates the class-overlapping problem. Ultimately, to enhance classification performance, it is essential not to generate a large number of new samples but to generate new samples in a targeted manner.

6. Conclusion

This paper proposes a noise-robust oversampling algorithm (NROMM) to address imbalanced classification problems in multi-class and mixed-type data. NROMM first addresses three major issues of existing model designs in the case of binary class and numerical attribute. To reduce the risk of noisy sample generation, NROMM adopts an advanced clustering algorithm to find arbitrary shapes for both the minority and majority classes. Then, NROMM develops a novel adaptive embedding method to generate samples tightly surrounding minority samples trapped in majorities. Besides, the samples that straddle class boundaries are selectively removed. To adapt to the mixed-type and multi-class scenarios, NROMM suggests choosing a dedicated metric to measure the distance between data samples and using generated samples recursively.

Experimental results on 20 benchmarks demonstrate the effectiveness and robustness of our algorithm. The algorithm significantly outperforms other oversampling baselines on most classifiers and assessment metrics. More precisely, NROMM outperforms the baselines at a large margin of around 10% on several benchmarks. Our algorithm also displays outstanding performance when dealing with a variety of data characteristics. Overall, this paper represents an encouraging step toward using oversampling to alleviate the multi-class and mixed-type imbalanced data problem. The dedicated noise-robust mechanisms in our algorithm can be flexibly integrated with the existing oversampling methods.

A potential weakness of NROMM is that it performs relatively poorly on datasets containing a large percentage of borderline samples, which is worth solving in the future. Besides, we plan to extend NROMM to graph-structured data, which is one of the most important data forms in reality.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary material

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.patcog.2022.109008](https://doi.org/10.1016/j.patcog.2022.109008)

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